

Aaryan Kainth

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WORK EXPERIENCE

Walt Disney Imagineering R&D (sites.disney.com/waltdisneyimagineering) Glendale, CA
Lab Associate June 2025 – September 2025

- Designed and fabricated an integrated remote control box to control a next-generation aquatic performance robot using SolidWorks, laser cutting, FDM 3D printing, and sheet metal bending
- Designed and constructed parts, tooling, and testing equipment for said robot, including mounts, adapters, a boat, and a lift to get the robot in and out of the water; utilized speed rail, waterjet cutting, and SLS 3D printing
- Designed and constructed an electronic, lightsaber-themed ping-pong tournament trophy within 24 hours
- Created a SolidWorks Hole Wizard preset to automatically create parameter-driven holes for heat set inserts for 3D prints
- Designed mounting points to integrate lighting and controller electronics into an onstage character prop

Cadense (cadense.com) Glendale, CA
R&D Design Engineering Intern September 2025 – Present

- Developed an adhesion system for rubber strips to add traction to the bottom of variable friction shoes
- Prototyping and testing various fall-prevention shoe designs to be put into future production

Hawkes Lab (hawkeslab.com) Goleta, CA
Undergraduate Research Assistant May 2023 – June 2025

- First authored a paper detailing the design and fabrication of a mounted anchor launcher that allows extraterrestrial rovers to traverse soft-sand slopes; presented it at the International Conference for Robotics and Automation in 2025
- Designed the clutch mechanism used in the anchor launcher
- Learned effective experimental design to create a model for anchor launch distance using Excel and MATLAB
- Optimized anchor force to weight ratio over design iterations, improving it by over 400%

Plume Design, Inc. (plume.com) Palo Alto, CA
Hardware Team Summer Intern June 2021 – August 2021

- Wrote Python testing scripts to quantify performance of 3 different kinds of product antennas
- Engineered a basic dipole antenna for 5 GHz bandwidth
- Learned how to use iBwave antenna simulation software, and trained additional team members on how to use it
- Leveraged Python and Raspberry Pi to collect and interpret data for wall mounted fall-detection prototype

PROJECTS

Capstone Project - Pinch-Free Animatronic Shells September 2025 – Present

- Connected relevant stakeholders at WDI R&D and UCSB's ME Capstone team to initiate capstone project collaboration
- Working in a team to develop thematic shells for various robot arm joints that prevent potentially hazardous pinch points
- Leveraging ROS2 Jazzy and Dynamixel MoveIt to animate the testbed, an OpenManipulator-X arm

CD Jukebox April 2025 – June 2025

- Designed a belt-driven lead screw system to adjust the linear position of a servo arm
- Wrote and tuned a PD controller in C++ to precisely drive the arm to within 2mm of target

BSA Troop 407 - Eagle Project February 2022 – June 2022

- Led 9 volunteers over the course of 103 man hours to build 3 ADA-compliant accessible animal tables for a local petting zoo, allowing them to better serve visitors who use wheelchairs

EDUCATION

University of California, Santa Barbara Goleta, CA
Bachelor of Science (B.S.) in Mechanical Engineering: 3.96 Cumulative GPA Expected June 2026
Master of Science (M.S.) in Mechanical Engineering: Beginning June 2026 Expected June 2027

SKILLS

Programming Languages: Python, MATLAB, C++

Manufacturing Skills: CAD, laser/waterjet cutting, 3D printing, mechanical design, rapid prototyping, soldering

Basic Proficiencies: Microsoft Office, Google Suite, LaTeX/Overleaf